TIME SERIES ANALYSIS

Dual Degree in Business Administration & Data and Business Analytics
BBADBA SEP-2026 TSA-NBDA.3.M.B

Area Mathematics
Number of sessions: 20
Academic year: 26-27
Degree course: THIRD
Number of credits: 3.0
Semester: 1º
Category: COMPULSORY
Language: English

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PhD in Computer and Telecommunications Engineering with experience in research, university teaching, and developing Artificial Intelligence solutions in the corporate environment. Currently working as a Data & AI Architect at Nationale-Nederlanden, where he participates in the design of data architectures, Artificial Intelligence platforms, and solutions based on Generative AI and language models. He combines this activity with university teaching, delivering courses related to Artificial Intelligence, Data Science, Machine Learning, and Programming, in addition to supervising Bachelor's theses and participating in evaluation committees. His teaching activity is oriented toward bringing professional experience into the classroom, integrating scientific foundations with real-world applications of Artificial Intelligence and Software Engineering.

Office Hours

Office hours will be on request. Please contact at:

Office hours are available on demand. Students should send an email to amoras@faculty.ie.edu in advance to schedule an appointment.

Office hours will be held after the in-person class or online (via Microsoft Teams) on Friday afternoons starting at 4:00 PM.

SUBJECT DESCRIPTION

Time series analysis comprises the methods used to analyse data indexed in time in order to extract meaningful statistics, identify structure, and produce forecasts. Time series analysis and forecasting are among the most widely applied data science techniques in business, finance, engineering and logistics. Examples of time series include the continuous monitoring of a person's heart rate, hourly readings of air temperature, the daily closing price of a company's stock, monthly rainfall, and yearly sales figures. The field has a well-established theoretical grounding in statistics and dynamic systems theory.

This course introduces the most important methods used to model time series data: decomposition, exponential smoothing and ARIMA models. All practical work is carried out in R, using the tidyverts family of packages (tsibble, feasts, fable). The course is taught as an applied engagement rather than a lecture series. Students work in teams of three to four, and each team is assigned one real business case at the start of the term. That case becomes their running problem for the whole semester: every technique introduced in class is immediately applied to it, evaluated on its merits, and either adopted or rejected with a documented justification.

PREREQUISITES FOR TAKING THIS SUBJECT:

- Fundamentals of Probability and Statistics.
- Fundamentals of Data Analysis.
- Programming and Data Visualisation

LEARNING OBJECTIVES

The main objective of this course is to provide students with a working methodology and a solid knowledge of the use of time series analysis in business. At the end of the course, students should be able to:

- Differentiate statistical models for independent observations from models for observations with temporal correlation.
- Identify patterns in correlated data: trend, seasonality and cycles.
- Understand and model time series data with exponential smoothing (ETS) and ARIMA models.
- Understand and apply the most important algorithms in the field.
- Design an appropriate methodology for each type of time series data (financial, hierarchical, seasonal, intermittent, multiple-seasonal).

- Use the standard diagnostic toolkit: ACF, PACF, CCF, stationarity tests, residual diagnostics and out-of-sample accuracy measures.

- Formulate a forecasting problem from an incomplete business brief, and defend a recommendation that is technically sound, properly validated and useful for the decision it serves.

TEACHING METHODOLOGY

The emphasis is on methods that are replicable and testable, and on practical examples. Sessions follow a fixed three-block structure:

- **30 minutes. Concept and R application.** A focused treatment of the technique of the week, oriented around what problem it solves, followed immediately by worked examples in R on a shared dataset.

- **30 minutes. Business Case Lab.** Each team applies the technique to its own assigned case. Working code is not the requirement; progress is. A valid output may be a plot, a methodological decision, a hypothesis, a data transformation, a fitted model, or an evidenced argument for why the technique does not apply to their case.

- **20 minutes. Case Clinic.** Selected teams present what they attempted, what they found and where they are blocked. The rest of the class challenges the reasoning, and the professor closes with feedback.

The classroom therefore functions as a laboratory: the substantive work on each case happens in class, with the professor present, rather than being deferred to homework. Because several teams work on each case, students see the same technique behave differently across comparable business problems. Each team maintains a decision log throughout the term, recording every assumption, data source, method tested and decision taken, with its justification. The log is the raw material of the final report.

Learning Activity	Weighting	Estimated time a student should dedicate to prepare for and participate in
Lectures	26.7 %	20.0 hours
Discussions	13.3 %	10.0 hours
Exercises in class, Asynchronous sessions, Field Work	20.0 %	15.0 hours
Group work	13.3 %	10.0 hours
Individual studying	26.7 %	20.0 hours
TOTAL	100.0 %	75.0 hours

AI POLICY

Restricted use of GenAI

In today's world, generative artificial intelligence (GenAI) is changing how we work, study and, in general, how we get things done. However, in the context of this course, the use of GenAI is not permitted, unless it is otherwise stated by the instructor. The use of GenAI tools would jeopardize the students' ability to acquire fundamental knowledge or skills of this course.

If a student is found to have used AI-generated content for any form of assessment, it will be considered academic misconduct, and the student might fail the respective assignment or the course.

BUSINESS CASES

The cases used in this course span different industries, data frequencies, decision horizons and failure modes, so that the same methodological syllabus produces four genuinely different analytical problems.

- CASE 1. HAPPY COW ICE CREAM (retail and FMCG) Asia Case Research Centre, The University of Hong Kong. A small vegan ice cream maker with transaction-level point-of-sale data from a campus store. Short history, weather- and calendar-driven demand, and a store manager who already intervenes in the series. Forecasting problem: short-horizon demand for replenishment and assortment decisions.

- CASE 2. BALEARIC AIRLINES (airlines, revenue management). IESE Business School. Daily demand for priority-class seats on a single Barcelona-Palma route with a two-tier fare structure and a fixed 186-seat capacity. Forecasting problem: setting a seat protection level two months before departure, which requires the demand distribution rather than a point forecast.

- CASE 3. CLEAN AIR LIMITED (manufacturing, supply chain) Ivey Publishing, Ivey Business School. Emission control systems for Indian commercial vehicles, with imported components on six- and twelve-month lead times committed a year in advance. Forecasting problem: an ordering and inventory plan under a short history containing structural breaks, with contractually asymmetric costs of error.

- CASE 4. WAYFAIR (e-commerce, forecasting at scale) Foresight: The International Journal of Applied Forecasting, 2025 Q4. A published account of Wayfair's hierarchical forecasting engine: millions of products, monthly demand one to eighteen months ahead, extreme sparsity and a rapidly expanding catalogue. Forecasting problem: whether hierarchical reconciliation and ensembling outperform forecasting each series independently, and at what cost in complexity.

PROGRAM

SESSION 1 (LIVE IN-PERSON)

Time series analysis compared with other domains of statistics

- Forecasting as a business problem: what can be forecast, and the three conditions that determine forecastability.
- Determining what to forecast: granularity, horizon, frequency and purpose.
- Forecasts, goals and plans, and the five-step forecasting workflow.
- The statistical perspective: forecast distributions, point forecasts and prediction intervals.
- Time series data in R: tsibbles, index, key and explicit gaps.
- Installation of R packages; assignment of teams and business cases.

R.A.: Demand Forecasting at Wayfair (FORESIGHT)

P.C.: Forecasting and revenue management at Balearic Airlines (I00007-PDF-ENG)

P.C.: Clean Air Limited: Forecasting for a Better Future (W42478-PDF-ENG)

P.C.: HAPPY COW ICE CREAM: DATA-DRIVEN SALES FORECASTING (HK1207-PDF-ENG)

P.C.: Marriott Rooms Forecasting (UV0353-PDF-ENG)

T.N.: Time Series (UV1301-PDF-ENG)

B.C.: Time Series Analysis and Forecasting (BEP505-PDF-ENG)

SESSION 2 (LIVE IN-PERSON)

Time series graphs

- Building a proper time plot: axis range, grid resolution and aspect ratio.
- Seasonal plots and seasonal subseries plots, including series with several seasonal periods.
- Additive versus multiplicative seasonal patterns, diagnosed from the plot.
- Scatterplots, non-linearities and the limits of the correlation coefficient.
- Lag plots as the bridge to autocorrelation.

SESSION 3 (LIVE IN-PERSON)

Autocorrelation

- Lagged variables: what they are, why they matter, and their relationship with scatterplots.
- The autocorrelation coefficient and the correlogram: formula, significance bounds and interpretation.
- Identifying trend and seasonal patterns from the ACF alone.
- White noise as the null hypothesis of forecasting, and as the target for model residuals.
- Partial autocorrelation, and what "partial" removes.

SESSION 4 (LIVE IN-PERSON)

Time series decomposition

- Additive versus multiplicative schemes.
- Moving averages and centred moving averages.
- Transformations to stabilise variance: logarithmic and Box-Cox, and choosing lambda defensibly.
- STL decomposition and the adjustment of the trend and season windows.
- Seasonally adjusted series and their use in business reporting.

SESSION 5 (LIVE IN-PERSON)

Benchmark methods and the general syntax of fitting models

- Naive, seasonal naive, mean and drift methods as the bar every later model must clear.
- Forecasting with decomposition.
- Training and test splits; the general model() and forecast() workflow in R.
- Residual diagnostics and forecast accuracy measures: MAE, RMSE, MAPE, MASE.
- Practical exercises with R.

SESSION 6 (LIVE IN-PERSON)

Simple exponential smoothing models

- Weighted averages of the past and the role of the smoothing parameter.
- Estimation by minimising the sum of squared errors.
- When a flat-forecast method is appropriate, and when it is not.
- Comparison against the session 5 benchmarks on the same test set.
- Practical exercises with R.

SESSION 7 (LIVE IN-PERSON)

Double and triple exponential smoothing models

- Holt's linear trend method, and the damped trend variant.
- Holt-Winters additive and multiplicative seasonal methods.
- Choosing between them from the patterns identified in Block 1.
- Practical exercises with R.

SESSION 8 (LIVE IN-PERSON)

Double and triple exponential smoothing models (continued)

- The ETS taxonomy: error, trend and seasonal components.
- Automatic model selection and the correct use of information criteria.
- Validation on the test set and residual diagnostics.
- Practical exercises with R.

SESSION 9 (LIVE IN-PERSON)

Intermediate test, and introduction to AR and MA models

- Assessment covering the 8 first sessions (20 % of the final grade).

- The shift from describing data to modelling its generating process.

SESSION 10 (LIVE IN-PERSON)

AR and MA models

- Autoregressive processes: today as a weighted sum of recent values.
- Moving average processes: today as a weighted sum of recent shocks.
- Reading the order of each from the ACF and PACF signatures.
- Practical exercises with R.

SESSION 11 (LIVE IN-PERSON)

ARMA models. Stationary time series, transformations and integration

- Stationarity: what it means and why every ARIMA model assumes it.
- Unit roots and formal tests.
- Differencing and seasonal differencing: how much of each is needed.
- Practical exercises with R.

SESSION 12 (LIVE IN-PERSON)

ARMA models. Stationary time series, transformations and integration (continued)

- Combining autoregressive and moving average terms.
- Model identification, estimation and residual diagnostics.
- The Ljung-Box test on residuals.
- Practical exercises with R.

SESSION 13 (LIVE IN-PERSON)

ARIMA models

- Non-seasonal ARIMA(p,d,q): specification and interpretation of orders.

- Order selection from the diagnostics, and comparison by information criteria.
- Forecasting and prediction intervals from a fitted ARIMA model.
- Practical exercises with R.

SESSION 14 (LIVE IN-PERSON)

Seasonal ARIMA (SARIMA) models

- Seasonal orders and the combined ARIMA(p,d,q)(P,D,Q)[m] specification.
- Identifying seasonal structure at the seasonal lags of the ACF and PACF.
- Practical exercises with R.

SESSION 15 (LIVE IN-PERSON)

Automatic selection of ARIMA models

- Stepwise search procedures and their limitations.
- The correct use of AICc, and why it cannot compare models with different differencing.
- Comparing ARIMA against ETS on out-of-sample accuracy.
- Practical exercises with R.

SESSION 16 (LIVE IN-PERSON)

Modelling time series with complex seasonality

- Multiple seasonal periods, Fourier terms and dynamic harmonic regression.
- Hierarchical and grouped time series, and forecast reconciliation.
- Intermittent and sparse demand: why the standard toolkit breaks down.
- Practical exercises with R.

SESSION 17 (LIVE IN-PERSON)

Integrated forecasting workshop

- The complete workflow applied end to end on each team's business case.
- Model comparison across families, on a common validation scheme.
- Selecting a final approach, and documenting the alternatives rejected.
- Practical exercises with R.

SESSION 18 (LIVE IN-PERSON)

Business case review and feedback

- Structured review of each team's methodology and supporting evidence.
- Translating forecast output into the business decision it serves: quantiles, service levels and the cost of each kind of error.
- Communicating uncertainty to a non-technical audience.
- Practical exercises with R.

SESSION 19 (LIVE IN-PERSON)

Group project presentations

- Each team presents its recommended forecasting approach, the alternatives considered and rejected, and the validation evidence supporting it (30 % of the final grade).
- Peer questioning and closing feedback.

SESSION 20 (LIVE IN-PERSON)

Final practical exam in R

- Assessment covering the full course (40 % of the final grade).
- Applied exercises in R: building a tsibble, diagnosing structure, fitting and validating models, and interpreting the result for a business decision.

EVALUATION CRITERIA

Disclaimer: The assessment method may be subject to modifications. Should changes occur, students will be notified on the first day of class and provided with a written document detailing the modifications.

IMPORTANT NOTE: MIDTERM RETAKE POLICY

The date for the midterm is the date of session 9, as specified in this syllabus. There is no retake for the midterm. If any student cannot attend the midterm for personal / work reasons the final exam will have a weight of 60% instead of 40%. That is, it will encompass both the midterm and the final exam weights.

criteria	percentage	Learning Objectives	Comments
Final Exam	30 %		
Group Work	20 %		
Class Participation	10 %		
Intermediate tests	30 %		
Individual work	10 %		

RE-SIT / RE-TAKE POLICY

Students with less than 4 out of 10 in the final exam will have to go to the retake exam, regardless of their performance on the other gradable items of the course.

In order to pass the course, you need a minimum grade of 3.5 in the final exam. If your grade in the final exam does not reach the threshold value of 3.5, you will fail the course, even in the case in which your weighted average (computed using the table above) exceeds 5.0. Each student has four chances to pass any given course distributed over two consecutive academic years: ordinary call exams and extraordinary call exams (re-sits) in June/July.

BIBLIOGRAPHY

Compulsory

- Hyndman & Athanasopoulos. (2018). *Forecasting: Principles and Practice.* OTexts. ISBN 9780987507112 (Digital)

- Robrt H. Sumway, David S. Stoffer.. (2019). *Time Series: A Data Analysis Approach Using R.* CRC PRESS. ISBN 9780367221096 (Digital)

SOFTWARE

- *R studio.* N/C.

- *Visual Studio Code.* N/C.

BEHAVIOR RULES

Please, check the University's Academic Integrity Policy here. The Program Director may provide further indications.

ATTENDANCE POLICY

Please, check the University's Attendance Policy here. The Program Director may provide further indications.

ETHICAL POLICY

Please, check the University's Student Code of conduct here. The Program Director may provide further indications.